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Department:
Basic Education
REPUBLIC OF SOUTH AFRICA

**NATIONAL
SENIOR CERTIFICATE
NASIONALE
SENIOR SERTIFIKAAT**

GRADE/GRAAD 12

**PHYSICAL SCIENCES: CHEMISTRY (P2)
FISIESE WETENSKAPPE: CHEMIE (V2)**

NOVEMBER 2025

MARKING GUIDELINES/NASIENRIGLYNE

MARKS/PUNTE: 150

**These marking guidelines consist of 22 pages.
Hierdie nasienriglyne bestaan uit 22 bladsye.**

QUESTION 1/VRAAG 1

- 1.1 C ✓✓ (2)
- 1.2 B ✓✓ (2)
- 1.3 A ✓✓ (2)
- 1.4 D ✓✓ (2)
- 1.5 A ✓✓ (2)
- 1.6 A ✓✓ (2)
- 1.7 A ✓✓ (2)
- 1.8 B ✓✓ (2)
- 1.9 C ✓✓ (2)
- 1.10 D ✓✓ (2)
- [20]**

QUESTION 2/VRAAG 2

- 2.1
- 2.1.1 D ✓ (1)
- 2.1.2 A and/en C ✓ (1)
- 2.1.3 E and/en F ✓ (1)

2.2

2.2.1

Marking criteria:

- Correct stem i.e. butanone. ✓
- Substituents (methyl) correctly identified. ✓
- IUPAC name is completely correct including numbering, sequence, hyphens and commas. ✓

Nasienkriteria:

- *Korrekte stam, d.i. butanoon.*
- *Substituente (metiel) korrek geïdentifiseer.*
- *IUPAC-naam heeltemal korrek insluitende nommering, volgorde, koppeltekens en kommas.*

3-methylbutan-2-one ✓✓✓/3-methylbutanone

3-metielbutan-2-oon/3-metielbutanoon

ACCEPT/AANVAAR

3-methyl-2-butanone/ methylbutanone/3-metiel-2-butanoon/metielbutanoon (3)

2.2.2

Marking criteria:

- Correct stem i.e. heptane. ✓
- Substituents (dichloro and dimethyl) correctly identified. ✓
- IUPAC name is completely correct including numbering, sequence, hyphens and commas. ✓

Nasienkriteria:

- Korrekte stam, d.i. heptaan.
- Substituente (dichloro en dimetiel) korrek geïdentifiseer.
- IUPAC-naam heeltemal korrek insluitende nommering, volgorde, koppeltekens en kommas.

2,6-dichloro-2,5-dimethylheptane ✓✓✓

2,6-dichloro-2,5-dimetielheptaan

(3)

2.2.3

Marking criteria:

- Correct stem i.e. hexene. ✓
- Substituents (ethyl and methyl) correctly identified. ✓
- IUPAC name is completely correct including numbering, sequence, hyphens and commas. ✓

Nasienkriteria:

- Korrekte stam, d.i. hekseen.
- Substituente (etiel en metiel) korrek geïdentifiseer.
- IUPAC-naam heeltemal korrek insluitende nommering, volgorde, koppeltekens en kommas.

3-ethyl-2-methylhex-2-ene ✓✓✓/3-ethyl-2-methyl-2-hexene

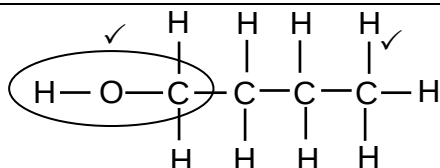
3-etiel-2-metielheks-2-een/3-etiel-2-metiel-2-hekseen

(3)

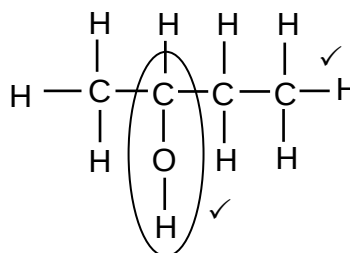
2.2.4

Marking criteria/Nasienkriteria:

- Hydroxyl group on the 1st C-atom. ✓
Hidroksiel groep op 1^{ste} C-atoom.
- Whole structure correct. ✓
Hele struktuur korrek.

**Marking criteria/Nasienkriteria:**

- Hydroxyl group on the 2nd C-atom. ✓
Hidroksiel groep op 2^{de} C-atoom.
- Whole structure correct. ✓
Hele struktuur korrek.

**IF/INDIEN**

- More than one functional group/wrong functional group: $0/2$ per molecule/molekule
Meer as een funksionele groep/foutiewe funksionele groep:
- Condensed structural formulae used
Gekondenseerde struktuurformules gebruik: Max/Maks. $2/4$
- Bond between O and H not shown, accept.
Binding tussen O en H nie gewys nie, aanvaar.

(4)

2.3.1 Combustion/oxidation/Verbranding/oksidasie ✓

(1)

2.3.2

Marking criteria:	Nasienkriteria:
(a) Volume O ₂ and C ₃ H ₈ used. ✓	(a) Volume O ₂ en C ₃ H ₈ gebruik. ✓
(b) Volume CO ₂ and H ₂ O produced ✓	(b) Volume CO ₂ en H ₂ O gevorm ✓
(c) Volume O ₂ remaining ✓	(c) Volume O ₂ wat oorbly ✓
(d) Addition of the three volumes ✓	(d) Optel van drie volumes ✓
(e) Correct final answer 66 cm ³ or 0,066 dm ³ ✓	(e) Korrekte finale antwoord 66 cm ³ of 0,066 dm ³ ✓

OPTION 1/OPSIE 1:

$V(\text{O}_2)_{\text{used/gebruik}} = 5V(\text{C}_3\text{H}_8)$
 $= (5)(8)$
 $= 40 \text{ cm}^3$ ✓ (a)

$V(\text{CO}_2)_{\text{formed/gevorm}} = 3V(\text{C}_3\text{H}_8)$
 $= (3)(8)$
 $= 24 \text{ cm}^3$

$V(\text{H}_2\text{O})_{\text{formed/gevorm}} = 4V(\text{C}_3\text{H}_8)$
 $= (4)(8)$
 $= 32 \text{ cm}^3$ ✓ (b)

$V(\text{O}_2)_{\text{unused}} = 50 - 40 = 10 \text{ cm}^3$ ✓ (c)

Volume gas in the container = $10 + 24 + 32$ ✓ (d)

Total volume gas in the container = 66 cm³ ✓ (e)

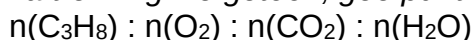
OPTION 2/ OPSIE 2:

	C ₃ H ₈	O ₂	CO ₂	H ₂ O
Initial volume (cm ³) Aanvanklike volume (cm ³)	8	50	0	0
Change volume (cm ³) Verandering in volume (cm ³) ✓ (a)	-8	-40	24	32
Final volume (cm ³) Finale volume (cm ³)	0	10 ✓ (c)	24	32
Total volume of gases in container (cm ³) Totale volume gas in houer (cm ³)		10 + 24 + 32 ✓ (d) = 66 cm ³ ✓ (e)		

ACCEPT/AANVAAR:

If subtraction not shown, give mark for the 10 cm³/

Indien aftrekking nie getoon, gee punt vir 10 cm³.



1 : 5 : 3 : 4

V =	8	40	24	32
	✓ (a)	✓ (b)		
			✓ (c)	
Total volume =	10 + 24 + 32 ✓ (d)			
	= 66 cm ³ ✓ (e)			

ACCEPT/ AANVAAR: Moles calculated using 22,4 dm ³ or any other molar gas volume. <i>Mol bereken deur 22,4 dm³ of ander molêre gasvolume te gebruik.</i>	
Marking criteria: (a) V(O ₂) and V(C ₃ H ₈)/ n(O ₂) and n(C ₃ H ₈) used. ✓ (b) V(CO ₂) and V(H ₂ O)/ n(CO ₂) and n(H ₂ O) produced. ✓ (c) V(O ₂)/ n(O ₂) remaining ✓ (d) Addition of the three volumes/moles ✓ (e) Correct final answer 66 cm ³ or 0,066 dm ³ ✓	Nasienkriteria: (a) V(O ₂) en V(C ₃ H ₈)/ n(O ₂) en n(C ₃ H ₈) gebruik. ✓ (b) V(CO ₂) en V(H ₂ O)/ n(CO ₂) en n(H ₂ O) gevorm. ✓ (c) V(O ₂)/ n(O ₂) wat oorbly ✓ (d) Optel van drie volumes/moles ✓ (e) Korrekte finale antwoord 66 cm ³ of 0,066 dm ³ ✓
$n(\text{C}_3\text{H}_8) = \frac{V}{V_m}$ $= \frac{0,008}{22,4}$ $= 3,571 \times 10^{-4} \text{ mol}$ $n(\text{O}_2)_{\text{change}} = (5)3,571 \times 10^{-4}$ $= 1,7855 \times 10^{-3} \text{ mol}$ $n(\text{O}_2)_{\text{change}} = \frac{V}{V_m}$ $1,7855 \times 10^{-3} = \frac{V}{22,4}$ $V(\text{O}_2)_{\text{change}} = 0,04 \text{ dm}^3$ $V(\text{O}_2)_{\text{remaining}} = 0,05 - 0,04$ $= 0,01 \text{ dm}^3 \quad \checkmark \text{ (c)}$ $n(\text{CO}_2) = (3) 3,571 \times 10^{-4}$ $= 1,0713 \times 10^{-3} \text{ mol}$ $n(\text{CO}_2) = \frac{V}{V_m}$ $1,0713 \times 10^{-3} = \frac{V}{22,4}$ $V(\text{CO}_2) = 0,024 \text{ dm}^3$ $n(\text{H}_2\text{O}) = (4)3,571 \times 10^{-4}$ $= 1,43 \times 10^{-3} \text{ mol}$ $n(\text{H}_2\text{O}) = \frac{V}{V_m}$ $1,43 \times 10^{-3} = \frac{V}{22,4}$ $V(\text{H}_2\text{O}) = 0,032 \text{ dm}^3$ $V_{\text{Total}} = 0,01 + 0,024 + 0,032 \quad \checkmark \text{ (d)}$ $= 0,066 \text{ dm}^3 \quad \checkmark \text{ (e)}$	$n(\text{C}_3\text{H}_8) = \frac{V}{V_m}$ $= \frac{0,008}{22,4}$ $= 3,571 \times 10^{-4} \text{ mol}$ $n(\text{O}_2)_{\text{change}} = (5)3,571 \times 10^{-4}$ $= 1,7855 \times 10^{-3} \text{ mol} \quad \checkmark \text{ (a)}$ $n(\text{O}_2)_{\text{ini}} = \frac{V}{V_m}$ $= \frac{0,05}{22,4}$ $= 2,232 \times 10^{-3} \text{ mol}$ $n(\text{O}_2)_{\text{remaining}} = 2,232 \times 10^{-3} - 1,7855 \times 10^{-3}$ $= 4,465 \times 10^{-4} \text{ mol} \quad \checkmark \text{ (c)}$ $n(\text{CO}_2) = (3)3,571 \times 10^{-4}$ $= 1,0713 \times 10^{-3} \text{ mol}$ $n(\text{H}_2\text{O}) = (4) \times 3,571 \times 10^{-4}$ $= 1,43 \times 10^{-3} \text{ mol} \quad \checkmark \text{ (b)}$ $n_{\text{Total}} = 4,465 \times 10^{-4} + 1,0713 \times 10^{-3} + 1,43 \times 10^{-3}$ $= 2,95 \times 10^{-3} \text{ mol} \quad \checkmark \text{ (d)}$ $n_{\text{Total}} = \frac{V}{V_m}$ $2,95 \times 10^{-3} = \frac{V}{22,4}$ $V_{\text{Total}} = 0,066 \text{ dm}^3 \quad \checkmark \text{ (e)}$

(5)
[22]

QUESTION 3/VRAAG 3

- 3.1 A series of organic compounds that can be described by the same general formula. ✓ (1 OR 0)

OR

A series of organic compounds in which one member differs from the next by a CH₂ group.

'n Reeks organiese verbindings wat deur dieselfde algemene formule beskryf kan word. (1 OF 0)

OR

'n Reeks organiese verbindings waarin die een lid van die volgende verskil met 'n CH₂-groep

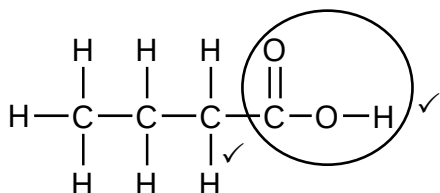
(1)

- 3.2 Ester ✓ and carboxylic acid ✓ / Ester en karboksielsuur

(2)

3.3

3.3.1

**Marking criteria/Nasienkriteria:**

(a) Functional group correct. ✓

Funksionele groep korrek.

(b) Whole structure correct. ✓

Hele struktuur korrek.

(2)

- 3.3.2 Methyl propanoate/Propyl methanoate/Ethyl ethanoate ✓✓ (2 OR/OF 0)
Metielpropanoaat/Propielmetanoaat/Etieletanoaat

(2)

- 3.4.1 Hydrogen bonds / Waterstofbindings ✓

(1)

- 3.4.2 Dipole-dipole forces / Dipool-dipoolkragte ✓

(1)

- 3.5 A ✓

The hydrogen bond is stronger than the dipole-dipole force. ✓

OR

The dipole-dipole force is weaker than the hydrogen bonds.

OR

Compound A has stronger intermolecular forces (than B).

OR

Compound B has weaker intermolecular forces (than A).

Die waterstofbinding is sterker as die dipool-dipoolkrag.

OR

Die dipool-dipoolkragte is swakker as die waterstofbinding.

OR

Verbinding A het sterker intermolekulêre kragte (as B).

OR

Verbinding B het swakker intermolekulêre kragte (as A).

(2)

- 3.6 Decreases/Afneem ✓

(1)

[12]

QUESTION 4/VRAAG 4

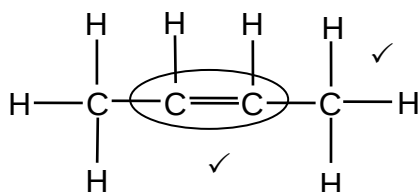
4.1

4.1.1 2-bromo✓butane✓/2-bromobutaan (2)

4.1.2 NaBr/Sodium bromide/Natriumbromied ✓ (1)

4.1.3 Addition/Addisie ✓
Hydrohalogenation/Hydrobromination/Hydrohalogenering/Hidrobrominering ✓ (2)4.1.4 (Concentrated) sulphuric acid/ H₂SO₄/Phosphoric acid/H₃PO₄/
(Gekonsentreerde) swaelsuur/Fosforsuur ✓ (1)

4.1.5

**Marking criteria/Nasienkriteria:**

(a) Correct functional group. ✓

Funksionele groep korrek.

(b) Whole structure correct. ✓

Hele struktuur korrek.

IF/INDIEN

- More than one functional group/wrong functional group:

- Meer as een funksionele groep/foutiewe funksionele groep: 0/2

- Correct condensed formula:

Korrekte gekondenseerde formule Max: 1/2 (2)

4.1.6 Concentrated strong base/Gekonsentreerde sterk basis ✓
Concentrated/Gekonsentreerde NaOH/KOH/LiOH (1)

4.2

4.2.1

Marking criteria/Nasienkriteria

If any one of the underlined key phrases in the **correct context** is omitted, deduct 1 mark./Indien enige van die onderstreepte frases in die **korrekte konteks** uitgelaat is, trek 1 punt af.

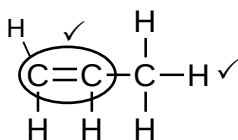
The underlined phrases must be in the correct context. / Die onderstreepte frases moet in die korrekte konteks wees.

The chemical process/reaction in which longer chain hydrocarbon/alkane molecules/ are broken down to shorter (more useful) molecules. ✓✓

Die chemiese proses/reaksie waarin langer kettingkoolwaterstof/alkaan- molekule afgebreek word in korter (meer bruikbare) molekules. (2)

4.2.2 Decolourisation/colour fades/becomes lighter in colour/colourless ✓
Ontkleuring/kleur raak dowwer/word ligter van kleur/kleurloos (1)

4.2.3

**Marking criteria/Nasienkriteria:**

- (a) Correct functional group. ✓
Funksionele groep korrek.
 (b) Whole structure correct. ✓
Hele struktuur korrek.

(2)

4.2.4 X / C₃H₆ / Propene / Propeen ✓

- X is unsaturated/has a double bond/is an alkene. ✓

ANY ONE

- X undergoes addition. ✓

OR

Alkenes are more reactive than alkanes/Unsaturated compounds react faster than saturated compounds.

Addition reaction is faster than substitution

Addition reaction does not need UV/light.

- X is onversadig/besit 'n dubbelbinding/is 'n alkeen.

ENIGE EEN

- X ondergaan addisie.

OF

Alkene is meer reaktief as alkane./Onversadigde verbindings reageer vinniger as versadigde verbindings.

Addisiereaksie is vinniger as substitusie.

Addisiereaksie benodig nie UV/lig.

(3)

[17]

QUESTION 5/VRAAG 5

5.1

NOTE/LET WELGive the mark for per unit time only if in context of reaction rate.Gee die punt vir per eenheid tyd slegs indien in konteks met reaksietempo.**ANY ONE:**

- Change in concentration ✓ of products/reactants per (unit) time. ✓
- Change in amount/number of moles/volume/mass of products or reactants per (unit) time.
- Amount/number of moles/volume/mass of products formed/reactants used per (unit) time.
- Rate of change in concentration/amount/number of moles/volume/mass. ✓✓ **(2 or 0)**

ENIGE EEN:

- Verandering in konsentrasie van produkte/reaktanse per (eenheid) tyd.
- Verandering in hoeveelheid/getal mol/volume/massa van produkte of reaktanse per (eenheid) tyd.
- Hoeveelheid/getal mol/volume/massa van produkte gevorm/reaktanse gebruik per (eenheid) tyd.
- Tempo van verandering in konsentrasie/ hoeveelheid/aantal mol/ volume/massa. **(2 of 0)**

(2)

5.2.1 10 (s) ✓

(1)

5.2.2

Marking criteria (a) Calculate the change: $[O_2]_{\text{final}} - [O_2]_{\text{initial}}$ / $n(O_2)_{\text{final}} - n(O_2)_{\text{initial}}$ / $n(CO_2)_{\text{final}} - n(CO_2)_{\text{initial}}$. ✓ Accept: 0,265 to 0,27 for $[O_2]_{\text{initial}}$ (b) Substitute 10 s in rate formula. ✓ (c) Multiply rate / concentration of O_2 by 3 dm^3. ✓ (d) USE mol ratio: $n(O_2) : n(CO_2) = 1 : 2$ ✓ (e) Final correct answer $= 0,072 \text{ (mol} \cdot \text{s}^{-1})$ ✓ RANGE: 0,06 – 0,072	Nasienkriteria: (a) Bereken die verandering: $[O_2]_{\text{finale}} - [O_2]_{\text{aanvank}}$ / $n(O_2)_{\text{finale}} - n(O_2)_{\text{aanvanklik}}$ / $n(CO_2)_{\text{finale}} - n(CO_2)_{\text{aanvanklik}}$ ✓ Aanvaar: 0,265 tot 0,27 vir $[O_2]_{\text{aanvank}}$ (b) Vervang 10 s in tempoformule. ✓ (f) Vermenigvuldig tempo/konsentrasie O_2 met 3 dm^3. ✓ (c) Gebruik molverhouding: $n(O_2) : n(CO_2) = 1 : 2$ ✓ (d) Finale korrekte antwoord $= 0,072 \text{ (mol} \cdot \text{s}^{-1})$ ✓ GEBIED: 0,06 – 0,072
OPTION 1/OPSIE 1: $\text{Rate/Tempo} = - \frac{\Delta c(O_2)}{\Delta t}$ $= - \left(\frac{0,15 - 0,27}{10 - 0} \right) \checkmark (a)$ $= 0,012 \text{ mol} \cdot \text{dm}^{-3} \cdot \text{s}^{-1} \checkmark (b)$ Rate/Tempo (O_2) in $\text{mol} \cdot \text{s}^{-1}$ $= cV$ $= (0,012)(3) \checkmark (c)$ $= 0,036 \text{ mol} \cdot \text{s}^{-1}$ Rate/Tempo (CO_2) = 2 x rate (O_2) $= 2 \times 0,036 \checkmark (d)$ $= 0,072 \text{ (mol} \cdot \text{s}^{-1}) \checkmark (e)$	OPTION 2/OPSIE 2: $\Delta c(O_2) = 0,27 - 0,15 \checkmark (a)$ $= 0,12 \text{ mol} \cdot \text{dm}^{-3}$ $\Delta n(O_2) = cV \downarrow$ $= 0,12(3) \checkmark (c)$ $= 0,36 \text{ mol}$ $\Delta n(CO_2) = 2n(O_2) \checkmark (d)$ $= 2(0,36)$ $= 0,72 \text{ mol}$ Rate(CO_2) = $\frac{\Delta n}{\Delta t}$ $= \frac{0,72}{10 - 0} \checkmark (b)$ $= 0,072 \text{ (mol} \cdot \text{s}^{-1}) \checkmark (e)$
OPTION 3/OPSIE 3: $n(O_2)_{\text{at } 0s} = cV$ $= 0,27 \times 3 \checkmark (c)$ $= 0,81 \text{ mol}$ $n(O_2)_{\text{at } 10s} = cV$ $= 0,15 \times 3$ $= 0,45 \text{ mol}$ Rate (O_2) = $-\frac{\Delta n}{\Delta t}$ $= - \left(\frac{0,45 - 0,81}{10 - 0} \right) \checkmark (a)$ $= 0,036 \text{ mol} \cdot \text{s}^{-1} \checkmark (b)$ Rate (CO_2) = 2 x rate (O_2) $= 2 \times 0,036 \checkmark (d)$ $= 0,072 \text{ (mol} \cdot \text{s}^{-1}) \checkmark (e)$	

(5)

5.2.3 O₂/Oxygen/Suurstof ✓ (1)

5.2.4 Increases/Neem toe ✓

Higher reaction rate./Concentration of reactants are higher. ✓
Hoër reaksietempo./Konsentrasie van die reaktanse is hoër.

ACCEPT/AANVAAR: pressure increased/druk verhoog. (2)

5.3.1 Absorption ✓
Absorpsie (1)

5.3.2 **Marking criteria/Nasienkriteria:**

If any one of the underlined key phrases in the **correct context** is omitted, deduct 1 mark./Indien enige van die onderstreepte frases in die **korrekte konteks** uitgelaat is, trek 1 punt af.

The underlined phrases must be in the correct context. / Die onderstreepte frases moet in die korrekte konteks wees.

Unstable (high energy) transition state from reactants to products. ✓✓
Onstabiele (hoë energie) oorgangs-toestand van reaktanse na produkte

ACCEPT/AANVAAR:

Unstable (high energy) transition state between reactants and products. ✓✓
Onstabiele (hoë energie) oorgangs-toestand tussen reaktanse en produkte. (2)

- 5.3.3
- Catalyst provides an alternative path with lower activation energy / lowers the activation energy. ✓
 - More particles have sufficient (kinetic) energy / kinetic energy greater (or equal to) activation energy. ✓
 - More effective collisions per unit time/second. ✓

OR

Higher frequency of effective collisions.

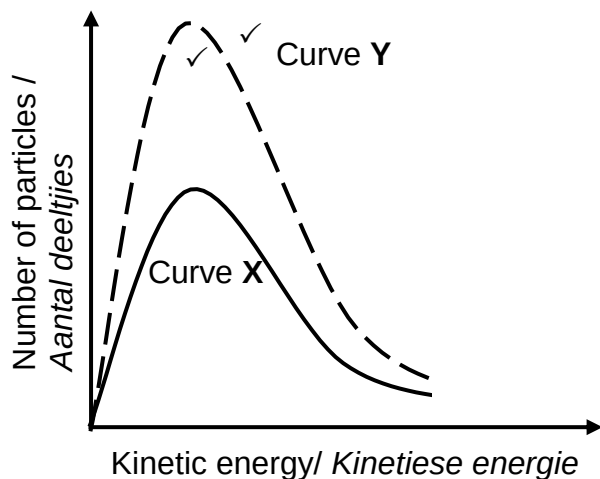
- Katalisator verskaf 'n alternatiewe roete met 'n laer aktiveringsenergie/Verlaag die aktiveringsenergie.
- Meer deeltjies het genoeg (kinetiese) energie/kinetiese energie groter (of gelyk aan) aktiveringsenergie.
- Meer effektiewe botsings per eenheid tyd/seconde.

OF

Hoër frekwensie van effektiewe botsings. (3)

5.3.4 Remains the same/Bly dieselfde ✓ (1)

5.3.5

**Marking criteria:**

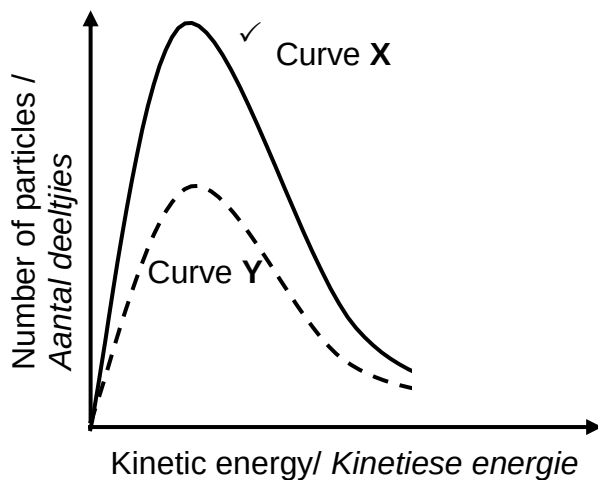
- Both curves start at origin and have correct shape with peaks at same E_k . ✓
- Peak of curve Y must be higher than curve X with peaks at same E_k . ✓

Nasienkriteria:

- Beide kurwes begin by die oorsprong en het dieselfde vorm met maksimums by dieselfde E_k .
- Maksimum van kurwe Y moet hoër wees as kurwe X met maksimums by dieselfde E_k .

IFIINDIEN:

- Both curves not labelled./ Beide kurwes nie benoem $0/2$
- Curves intersect at any other point, beside the origin. Kurwes kruis by enige ander punt as oorsprong. Max: $1/2$.

IFIINDIEN:Max/Maks: $1/2$ (2)
[20]

QUESTION 6/VRAAG 6

6.1

Marking criteria/Nasienkriteria:

6.1.1

If any one of the underlined key phrases in the **correct context** is omitted, deduct 1 mark. / Indien enige van die onderstreepte frases in die **korrekte konteks** uitgelaat is, trek 1 punt af.

The underlined phrases must be in the correct context. / Die onderstreepte frases moet in die korrekte konteks wees.

When the equilibrium in a closed system is disturbed, the system will re-instate a new equilibrium by favouring the reaction that will cancel/oppose the disturbance. ✓✓

Wanneer die ewewig in 'n geslote sisteem versteur word, sal die sisteem 'n nuwe ewewig instel deur die reaksie te bevoordeel wat die versteuring kanselleer/teenwerk. (2)

6.1.2

(Mass) decreases. / (Massa) Afneem. ✓ (1)

6.1.3

- Decrease in amount of OH^- ions / concentration of OH^- ions, favours the reaction that increases the amount/concentration of OH^- ions. ✓ **OR** Acid/ HCl/H^+ reacts with OH^- ions.

- The forward reaction is favoured ✓ **OR** The amount/concentration of the products increases.

- 'n Afname in hoeveelheid OH^- -ione / konsentrasie OH^- -ione bevoordeel die reaksie wat die hoeveelheid/konsentrasie van OH^- -ione laat toeneem. **OF**

Suur / HCl/H^+ reageer met OH^- ione.

- Die voorwaartse reaksie is bevoordeel. **OF** Die hoeveelheid/konsentrasie van die produkte neem toe. (2)

6.2.1

Endothermic/Endotermies ✓ (1)

6.2.2

- With an increase in the temperature the K_c value increases. ✓
- The concentration of the products increases. **OR** Concentration of reactants decreases. **OR** The forward reaction is favoured. ✓
- (According to Le Chatelier's principle) an increase in temperature favours the endothermic reaction. ✓

- Met 'n toename in temperatuur neem die K_c -waarde toe.
- Die konsentrasie van die produkte neem toe. **OF** Konsentrasie van die reaktante neem af. **OF** Voorwaartse reaksie is bevoordeel.
- (Volgens Le Chatelier se beginsel) sal 'n toename in temperatuur die endotermiese reaksie bevoordeel. (3)

6.2.3

CALCULATIONS USING MOLES**BEREKENINGE WAT MOL GEBRUIK****Marking criteria:**

- (a) Calculate number of moles NH_4HS ($\frac{70}{51}$) ✓ **OR** 1,37 moles
- (b) **USING RATIO:** $\text{NH}_4\text{HS} : \text{NH}_3 : \text{H}_2\text{S} : = 1 : 1 : 1$ ✓
- (c) Calculate $c(\text{NH}_3)$ and $c(\text{H}_2\text{S})$ at equilibrium (divide equilibrium moles by 3) ✓
- (d) Correct K_c expression ✓
- (e) Substitute $K_c = 18 \times 10^{-2}$ ✓
- (f) $n(\text{NH}_4\text{HS})_{\text{eq}} = n(\text{NH}_4\text{HS})_{\text{in}} - n(\text{NH}_4\text{HS})_{\text{change}}$ **OR**
 $m(\text{NH}_4\text{HS})_{\text{eq}} = m(\text{NH}_4\text{HS})_{\text{in}} - m(\text{NH}_4\text{HS})_{\text{change}}$ ✓
- (g) Substitute 51 in $n = \frac{m}{M}$ ✓
- (h) **CORRECT** final answer: $m = 5,61 \text{ g}$ ✓
 Range: 4,96 – 5,74 g

Nasienkriteria:

- (a) Bereken aantal mol NH_4HS ($\frac{70}{51}$) ✓ **OF** 1,37 mol
- (b) **GEBRUIK VERHOUDING:** $\text{NH}_4\text{HS} : \text{NH}_3 : \text{H}_2\text{S} : = 1 : 1 : 1$ ✓
- (c) Bereken $c(\text{NH}_3)$ en $c(\text{H}_2\text{S})$ by ewewig (deel ewewig mol met 3) ✓
- (d) Korrekte K_c uitdrukking ✓
- (e) Vervang $K_c = 18 \times 10^{-2}$ ✓
- (f) $n(\text{NH}_4\text{HS})_{\text{eq}} = n(\text{NH}_4\text{HS})_{\text{in}} - n(\text{NH}_4\text{HS})_{\text{change}}$ **OF**
 $m(\text{NH}_4\text{HS})_{\text{eq}} = m(\text{NH}_4\text{HS})_{\text{in}} - m(\text{NH}_4\text{HS})_{\text{change}}$ ✓
- (g) Vervang 51 in $n = \frac{m}{M}$ ✓
- (h) **KORREKTE** finale antwoord: $m = 5,61 \text{ g}$ ✓
 Gebied: 4,95 – 5,74 g

OPTION 1/OPSIE 1:

$$n = \frac{m}{M}$$

$$= \frac{70}{51} \quad \checkmark \text{ (a)}$$

	$\text{NH}_4\text{HS}(\text{s})$	$\text{NH}_3(\text{g})$	$\text{H}_2\text{S}(\text{g})$	
Initial amount (mol) Aanvanklike hoeveelheid (mol)	1,37	0	0	
Change (mol) Verandering (mol)	x	x	x	✓ (b)
Equilibrium amount (mol) Ewewig hoeveelheid (mol)	1,37 - x	x	x	
Equilibrium concentration ($\text{mol} \cdot \text{dm}^{-3}$) Ewewigkonsentrasie ($\text{mol} \cdot \text{dm}^{-3}$)		$\frac{x}{3}$	$\frac{x}{3}$	✓ (c)

$$K_c = [\text{NH}_3][\text{H}_2\text{S}] \quad \checkmark \text{ (d)}$$

$$\checkmark \text{ (e)} \quad 18 \times 10^{-2} = \left(\frac{x}{3}\right)^2$$

$$x = 1,27$$

$$n(\text{NH}_4\text{HS})_{\text{eq}} = 1,37 - 1,27 \quad \checkmark \text{ (f)}$$

$$= 0,1 \text{ mol}$$

$$m(\text{NH}_4\text{HS})_{\text{eq}} = nM$$

$$= 0,1 \times 51 \quad \checkmark \text{ (g)}$$

$$= 5,1 \text{ g} \quad \checkmark \text{ (h)}$$

No K_c expression, correct substitution
 Geen K_c - uitdrukking, korrekte substitusie:
 Max./Maks. 7/8

Wrong K_c expression/
 Verkeerde K_c -uitdrukking: Max./Maks. 6/8

OPTION 2/OPSIE 2:

$$n = \frac{m}{M}$$

$$= \frac{70}{51} \checkmark (a)$$

$$K_c = [\text{NH}_3][\text{H}_2\text{S}] \checkmark (d)$$

$$\checkmark (e) 18 \times 10^{-2} = x^2$$

$$x = 0,42$$

	NH ₄ HS(s)	NH ₃ (g)	H ₂ S(g)	
Initial amount (mol) Aanvanklike hoeveelheid (mol)	1,37	0	0	
Change (mol) Verandering (mol)	1,26	1,26	1,26	✓ (b)
Equilibrium amount (mol) Ewewig hoeveelheid (mol)	✓ (f) 0,11	1,26	1,26	
Equilibrium concentration (mol·dm ⁻³) Ewewigkonsentrasie (mol·dm ⁻³)		0,42	0,42	✓ (c)

$$m(\text{NH}_4\text{HS})_{\text{eq}} = nM$$

$$= 0,11 \times 51 \checkmark (g)$$

$$= 5,61 \text{ g} \checkmark (h)$$

OR/OF

$$m(\text{NH}_4\text{HS})_{\text{change}} = nM$$

$$= 1,26 \times 51 \checkmark (g)$$

$$= 64,26 \text{ g}$$

$$m(\text{NH}_4\text{HS})_{\text{eq}} = 70 - 64,26 \checkmark (f)$$

$$= 5,74 \text{ g} \checkmark (h)$$

No K_c expression, correct substitution
 Geen K_c- uitdrukking, korrekte substitusie:
 Max./Maks. 7/8

Wrong K_c expression/
 Verkeerde K_c-uitdrukking: Max./Maks. 6/8

CALCULATIONS USING CONCENTRATION
BEREKENINGE WAT KONSENTRASIE GEBRUIK

	NH ₄ HS(s)	NH ₃ (g)	H ₂ S(g)
Initial concentration (mol·dm ⁻³) Aanvanklike konsentrasie (mol·dm ⁻³)		0	0
Change (mol·dm ⁻³) Verandering (mol·dm ⁻³)		✓(b) x	x
Equilibrium concentration (mol·dm ⁻³) Ewewigskonsentrasie (mol·dm ⁻³)		x	x

$$\begin{aligned} \checkmark(d) K_c &= [\text{NH}_3][\text{H}_2\text{S}] \\ \checkmark(e) 18 \times 10^{-2} &= x^2 \\ x &= 0,424 \text{ mol}\cdot\text{dm}^{-3} \end{aligned}$$

$$\begin{aligned} n\text{NH}_3(\text{change}) &= cV \\ &= (0,424)(3) \checkmark(c) \\ &= 1,272 \text{ mol} \end{aligned}$$

$$n(\text{NH}_4\text{HS})_{\text{change}} = n\text{NH}_3(\text{change}) = 1,272 \text{ mol}$$

$$\begin{aligned} \checkmark(a) \\ n(\text{NH}_4\text{HS})_{\text{eq}} &= \frac{70}{51} - 1,272 \checkmark(f) \\ &= 0,098 \text{ mol} \end{aligned}$$

$$\begin{aligned} m(\text{NH}_4\text{HS})_{\text{eq}} &= nM \\ &= 0,098 \times 51 \checkmark(g) \\ &= 4,998 \text{ g} \checkmark(h) \end{aligned}$$

OR/OF

$$\begin{aligned} m(\text{NH}_4\text{HS})_{\text{change}} &= nM \\ &= 1,272 \times 51 \checkmark(g) \\ &= 64,872 \text{ g} \end{aligned}$$

$$\begin{aligned} m(\text{NH}_4\text{HS})_{\text{eq}} &= 70 - 64,872 \checkmark(f) \\ &= 5,128 \text{ g} \checkmark(h) \end{aligned}$$

No K_c expression, correct substitution
 Geen K_c- uitdrukking, korrekte substitusie: Max./Maks. 7/8

Wrong K_c expression/
 Verkeerde K_c-uitdrukking: Max./Maks. 6/8

(8)
 [17]

QUESTION 7/VRAAG 7

7.1.1 H_2PO_4^- ✓
 K_a of H_2PO_4^- greater/higher than K_a of HPO_4^{2-} ✓
Accept: K_a of H_2PO_4^- greater (2)

7.1.2 HPO_4^{2-} ✓ (1)

7.1.3 H_2PO_4^- **OR/OF** HPO_4^{2-} ✓ (1)

7.1.4 Basic/Basies ✓ (1)

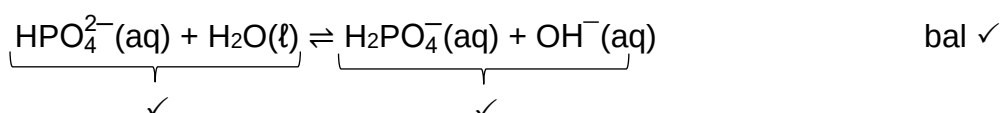
7.1.5

Marking criteria/Nasienkriteria:

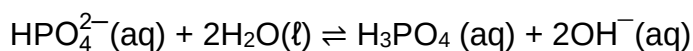
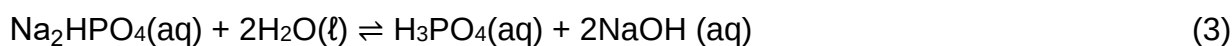
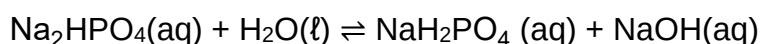
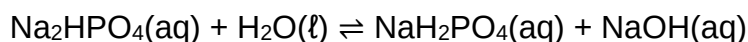
- Reactants ✓ Products ✓ Balancing ✓
 Reaktanse ✓ Produkte ✓ Balansering ✓

- Ignore/Ignoreer → and phases/en fases

Marking rule 6.3.10/Nasienreël 6.3.10



OR/OF

**ACCEPT/AANVAAR:**

7.2.1

Marking criteria:	Nasienkriteria:
a) Any formula: $\text{pH} = -\log[\text{H}_3\text{O}^+]$ / $\text{pH} = -\log[\text{H}^+] / [\text{H}_3\text{O}^+] = 10^{-\text{pH}}$ / $\text{pOH} = -\log[\text{OH}^-]$ / $[\text{H}_3\text{O}^+][\text{OH}^-] = 10^{-14}$ / $\text{pH} + \text{pOH} = 14$ ✓ b) Substitute 12,62 in $\text{pH} = -\log[\text{H}_3\text{O}^+]$ / $\text{pH} + \text{pOH} = 14$ ✓ c) Substitute calculated $[\text{H}_3\text{O}^+]$ in $[\text{H}_3\text{O}^+][\text{OH}^-]$ / 1,38 in $\text{pOH} = -\log[\text{OH}^-]$ ✓ d) Final answer: $0,04 \text{ mol} \cdot \text{dm}^{-3}$ ✓ RANGE: $0,04 - 0,042 \text{ mol} \cdot \text{dm}^{-3}$	a) Enige formule: $\text{pH} = -\log[\text{H}_3\text{O}^+]$ / $\text{pH} = -\log[\text{H}^+] / [\text{H}_3\text{O}^+] = 10^{-\text{pH}}$ / $\text{pOH} = -\log[\text{OH}^-]$ / $[\text{H}_3\text{O}^+][\text{OH}^-] = 10^{-14}$ / $\text{pH} + \text{pOH} = 14$ ✓ b) Vervang 12,62 in $\text{pH} = -\log[\text{H}_3\text{O}^+]$ / $\text{pH} + \text{pOH} = 14$ ✓ c) Vervang berekende $[\text{H}_3\text{O}^+]$ in $[\text{H}_3\text{O}^+][\text{OH}^-]$ / 1,38 in $\text{pOH} = -\log[\text{OH}^-]$ ✓ d) Finale antwoord: $0,04 \text{ mol} \cdot \text{dm}^{-3}$ ✓ GEBIED: $0,04 - 0,042 \text{ mol} \cdot \text{dm}^{-3}$
OPTION 1/OPSIE 1 $\text{pH} = -\log[\text{H}_3\text{O}^+]$ 12,62 ✓(b) $= -\log[\text{H}_3\text{O}^+]$ OR/OF $[\text{H}_3\text{O}^+] = 10^{-12,62}$ Any one/Enige een ✓(a) $[\text{H}_3\text{O}^+] = 2,4 \times 10^{-13}$ $[\text{H}_3\text{O}^+][\text{OH}^-] = 10^{-14}$ ✓(c) $(2,4 \times 10^{-13})[\text{OH}^-] = 1 \times 10^{-14}$ $[\text{OH}^-] = 0,0417 \text{ mol} \cdot \text{dm}^{-3}$ ✓(d) (0,04)	
OPTION 2/OPSIE 2 $\text{pH} + \text{pOH} = 14$ ✓(b) $12,62 + \text{pOH} = 14$ Any one/Enige een ✓(a) $\text{pOH} = 1,38$ $\text{pOH} = -\log[\text{OH}^-]$ 1,38 ✓(c) $= -\log[\text{OH}^-]$ $[\text{OH}^-] = 0,042 \text{ mol} \cdot \text{dm}^{-3}$ ✓(d) (0,04)	

(4)

7.2.2

**POSITIVE MARKING FROM QUESTION 7.2.1/
POSITIEWE NASIEN VANAF VRAAG 7.2.1**

Marking criteria (a) Substitute: $0,2 \text{ mol} \cdot \text{dm}^{-3}$ AND $0,015 \text{ dm}^3$ in $n = cV$ ✓ (b) USING RATIO: $n(\text{OH}^-) : n\text{HCl} = 1 : 1 / n(\text{Ba}(\text{OH})_2) :$ $n(\text{HCl}) = 1 : 2$ ✓ (c) Substitute: $c(\text{OH}^-)$ AND $0,04$ in $n = cV$ ✓ (d) Calculate $n(\text{OH}^-)_{\text{ini}}$ $= n(\text{OH}^-)_{\text{reacted with HCl}} + n(\text{OH}^-)_{\text{fin}}$ ✓✓ (e) USING RATIO: $n(\text{Ba}(\text{OH})_2) : n(\text{OH}^-) = 1 : 2$ ✓ (f) Multiply $n\text{Ba}(\text{OH})_2$ in 25 cm^3 by 4 OR Divide by $0,025 \text{ dm}^3$ AND multiply by $0,1 \text{ dm}^3$ ✓ (g) Final correct answer: $9,34 \times 10^{-3} \text{ mol}$ ✓ Range: $9,2 \times 10^{-3} - 9,36 \times 10^{-3}$	Nasienkriteria: (a) Vervang: $0,2 \text{ mol} \cdot \text{dm}^{-3}$ EN $0,015 \text{ dm}^3$ in $n = cV$ ✓ (b) GEBRUIK VERHOUDING: $n(\text{OH}^-) : n\text{HCl} = 1 : 1 / n(\text{Ba}(\text{OH})_2) :$ $n(\text{HCl}) = 1 : 2$ ✓ (c) Vervang: $c(\text{OH}^-)$ EN $0,04$ in $n = cV$ ✓ (d) Bereken $n(\text{OH}^-)_{\text{ini}}$ $= n(\text{OH}^-)_{\text{reageer met HCl}} + n(\text{OH}^-)_{\text{finale}}$ ✓✓ (e) GEBRUIK VERHOUDING: $n(\text{Ba}(\text{OH})_2) : n(\text{OH}^-) = 1 : 2$ ✓ (f) Vermenigvuldig $n\text{Ba}(\text{OH})_2$ in 25 cm^3 met 4 OF Deel deur $0,025 \text{ dm}^3$ EN vermenigvuldig met $0,1 \text{ dm}^3$ (g) Finale korrekte antwoord: $9,34 \times 10^{-3} \text{ mol}$ ✓ Gebied: $9,2 \times 10^{-3} - 9,36 \times 10^{-3}$
OPTION 1/OPSIE 1 $n\text{HCl} = cV$ $= (0,2)(0,015)$ ✓(a) $= 3 \times 10^{-3} \text{ mol}$ $n\text{OH}^- = n\text{HCl}$ ✓(b) $= 3 \times 10^{-3} \text{ mol}$ $n\text{OH}^-_{\text{final}} = cV$ $= (0,0417)(0,04)$ ✓(c) $= 1,67 \times 10^{-3} \text{ mol}$ $n\text{OH}^-_{\text{ini}} = 3 \times 10^{-3} + 1,67 \times 10^{-3}$ ✓✓(d) $= 4,67 \times 10^{-3} \text{ mol}$ $n\text{Ba}(\text{OH})_2 = \frac{1}{2} n\text{OH}^-_{\text{ini}}$ $= \frac{1}{2} (4,67 \times 10^{-3})$ ✓(e) $= 2,33 \times 10^{-3} \text{ mol in } 25 \text{ cm}^3$ In $100 \text{ cm}^3 = (4)(2,33 \times 10^{-3})$ ✓(f) $= 9,34 \times 10^{-3} \text{ mol}$ ✓(g)	OPTION 2/OPSIE 2 $n\text{HCl} = cV$ $= (0,2)(0,015)$ ✓(a) $= 3 \times 10^{-3} \text{ mol}$ $n\text{Ba}(\text{OH})_2 = \frac{1}{2} n\text{HCl}$ $= \frac{1}{2} (3 \times 10^{-3})$ ✓(b) $= 1,5 \times 10^{-3} \text{ mol}$ $n\text{OH}^-_{\text{final}} = cV$ $= (0,0417)(0,04)$ ✓(c) $= 1,67 \times 10^{-3} \text{ mol}$ $n\text{Ba}(\text{OH})_2 = \frac{1}{2} n\text{OH}^-_{\text{ini}}$ $= \frac{1}{2} (1,67 \times 10^{-3})$ ✓(e) $= 8,33 \times 10^{-4} \text{ mol}$ $n\text{Ba}(\text{OH})_2_{\text{ini}} = 1,5 \times 10^{-3} + 8,33 \times 10^{-4}$ ✓✓(d) $= 2,33 \times 10^{-3} \text{ mol in } 25 \text{ cm}^3$ In $100 \text{ cm}^3 = (4)(2,33 \times 10^{-3})$ ✓(f) $= 9,34 \times 10^{-3} \text{ mol}$ ✓(g)

OPTION 3/OPSIE 3:

$$n\text{HCl} = cV$$

$$= (0,2)(0,015) \checkmark \text{(a)}$$

$$= 3 \times 10^{-3} \text{ mol}$$

$$n\text{Ba(OH)}_2 = \frac{1}{2} n\text{HCl}$$

$$= \frac{1}{2} (3 \times 10^{-3}) \checkmark \text{(b)}$$

$$= 1,5 \times 10^{-3} \text{ mol}$$

$$c\text{Ba(OH)}_2 = \frac{1}{2} c\text{OH}^-_{\text{final}} \checkmark \text{(e)}$$

$$= \frac{1}{2} (0,0417)$$

$$= 0,02085 \text{ mol} \cdot \text{dm}^{-3}$$

$$n\text{Ba(OH)}_2 = cV$$

$$= (0,02085 \times 0,04) \checkmark \text{(c)}$$

$$= 8,34 \times 10^{-4} \text{ mol}$$

$$n\text{Ba(OH)}_{2 \text{ ini}} = 1,5 \times 10^{-3} + 8,34 \times 10^{-4} \checkmark \checkmark \text{(d)}$$

$$= 2,34 \times 10^{-3} \text{ mol in } 25 \text{ cm}^3$$

$$\text{In } 100 \text{ cm}^3 = (4)(2,34 \times 10^{-3}) \checkmark \text{(f)}$$

$$= 9,36 \times 10^{-3} \text{ mol} \checkmark \text{(g)}$$

✓(g) in all options/lin alle opsies:

$$n\text{Ba(OH)}_2 \text{ in } 25 \text{ cm}^3 = cV$$

$$2,33 \times 10^{-3} \text{ mol} = c(0,025)$$

$$c\text{Ba(OH)}_2 \text{ in } 25 \text{ cm}^3 = 0,0936 \text{ mol} \cdot \text{dm}^{-3}$$

$$n\text{Ba(OH)}_2 \text{ in } 100 \text{ cm}^3 = cV$$

$$= (0,0936)(0,1)$$

$$= 0,0936 \text{ mol}$$

(8)
[20]**QUESTION 8/VRAAG 8****8.1 ANY ONE:**

- A substance whose (aqueous) solution contains ions. ✓✓ (2 OR/OF 0)
- Substance that dissolves in water to give a solution that conducts electricity (through movement of ions).

OR

- A substance that dissociates to form ions in water/in molten state.

ENIGE EEN:

- 'n Stof waarvan die (waterige) oplossing ione bevat.
- 'n Stof wat in water oplos om 'n oplossing te vorm wat elektrisiteit gele (deur die beweging van ione).

OF

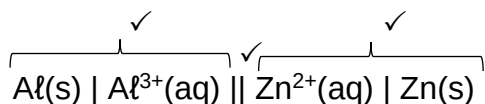
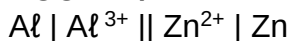
- 'n Stof wat dissosieer om ione in water te vorm/in gesmelte toestand

(2)

8.2 Al³⁺/Aluminium ion/ioon ✓Al is oxidised **OR** Al is a stronger reducing agent. ✓Al is geoksideer **OF** Al is 'n sterker reduseermiddel.

(2)

8.3

**OR/OF****ACCEPT/AANVAAR**

(3)

8.4

Marking criteria (a) Calculate $n(\text{Al}^{3+})$ ✓ (b) USE mol ratio: $n\text{Al}_2(\text{SO}_4)_3 : n\text{Al}^{3+} = 1 : 2$ ✓ (c) Multiply $n\text{Al}_2(\text{SO}_4)_3$ by $M = 342$ ✓ (d) Final correct answer $= 42,75 \text{ g}$ ✓	Nasienkriteria: (a) Bereken $n(\text{Al}^{3+})$ ✓ (b) Gebruik molverhouding: $n\text{Al}_2(\text{SO}_4)_3 : n\text{Al}^{3+} = 1 : 2$ ✓ (c) Vermenigvuldig $n\text{Al}_2(\text{SO}_4)_3$ met $M = 342$ ✓ (d) Finale korrekte antwoord $= 42,75 \text{ g}$ ✓
OPTION 1/OPSIE 1: $n(\text{Al}^{3+}) = cV$ $= \boxed{(1)(0,25)}$ ✓ (a) $= 0,25 \text{ mol}$ $n\text{Al}_2(\text{SO}_4)_3 = \frac{1}{2} n(\text{Al}^{3+})$ $= \frac{1}{2} (0,25)$ ✓ (b) $= 0,125 \text{ mol}$ $n\text{Al}_2(\text{SO}_4)_3 = \frac{m}{M}$ $0,125 = \frac{m}{342}$ ✓ (c) $m = 42,75 \text{ g}$ ✓ (d)	OPTION 2/OPSIE 2: $c = \frac{m}{VM}$ $\checkmark (a) \quad 1 = \frac{m}{(0,25)(342)}$ ✓ (c) $m = 85,5 \text{ g}$ $\text{Al}_2(\text{SO}_4)_3 \rightarrow 2 \text{ mol} \cdot \text{dm}^{-3} \text{ Al}^{3+}$ $m(\text{Al}_2(\text{SO}_4)_3) = \frac{1}{2} (85,5)$ ✓ (b) $= 42,75 \text{ g}$ ✓ (d)

(4)
[11]

QUESTION 9/VRAAG 9

9.1 Electrolytic/Elektrolitiese ✓

(1)

9.2.1 Increases/Neem toe ✓

(1)

9.2.2 Decrease ✓

- More copper (II) ions/ Cu^{2+} are reduced than formed./ Only copper (II) ions/ Cu^{2+} are reduced. ✓
- Copper (II) ion/ Cu^{2+} is a stronger oxidising agent than the zinc (II) ion/ Zn^{2+} ✓

ORZinc (II) ion/ Zn^{2+} is a weaker oxidising agent than the copper (II) ion/ Cu^{2+} *Afneem*

- Meer koper(II)ione/ Cu^{2+} word gereduseer as wat gevorm word./ Slegs koper(II)ione/ Cu^{2+} word gereduseer.
- Koper(II)ioon/ Cu^{2+} is 'n sterker oksideermiddel as die sink(II)ioon/ Zn^{2+}

OFSink(II)ioon/ Zn^{2+} is 'n swakker oksideermiddel as die koper(II)ioon/ Cu^{2+}

(3)

9.2.3

Marking criteria	Nasienkriteria:
(a) USING RATIO: $n(\text{Cu}^{2+}) : n(\text{Zn}^{2+}) = 1 : 1$ ✓	(a) GEBRUIK VERHOUDING ✓ $n(\text{Cu}^{2+}) : n(\text{Zn}^{2+}) = 1 : 1$
(b) Formula: $n = \frac{m}{M}$ ✓	(b) Formule: $n = \frac{m}{M}$ ✓
(c) Substitute 65 AND $n(\text{Zn}^{2+})$ in $n = \frac{m}{M}$ ✓	(c) Vervang 65 EN $n(\text{Zn}^{2+})$ in $n = \frac{m}{M}$ ✓
(d) Subtraction of moles ✓	(d) Aftrek van aantal mol ✓
(e) Substitute 63,5 AND $n(\text{Cu})$ in $n = \frac{m}{M}$ ✓	(e) Vervang 63,5 EN $n(\text{Cu})$ in $n = \frac{m}{M}$ ✓
(f) Final correct answer: <u>9,6 g</u> ✓ (NO RANGE)	(f) Finale korrekte antwoord: <u>9,6 g</u> ✓ (GEEN GEBIED)

$$n(\text{Zn}^{2+}) = n(\text{Cu}^{2+})$$

$$= 0,05 \text{ mol} \quad \checkmark \text{(a)}$$

$$m(\text{Zn}) = nM \quad \checkmark \text{(b)}$$

$$= (0,05)(65) \quad \checkmark \text{(c)}$$

$$= 3,25 \text{ g Zn from R}$$

$$0,15 - 0,05 \quad \checkmark \text{(d)} = 0,1 \text{ mol Cu}$$

$$m(\text{Cu}) = nM$$

$$= (0,1)(63,5) \quad \checkmark \text{(e)}$$

$$= 6,35 \text{ g Cu from R}$$

$$\text{Change in mass} = 6,35 + 3,25$$

$$= 9,6 \text{ g} \quad \checkmark \text{(f)} \quad (\text{decrease in mass})$$

(6)
[11]**TOTAL/TOTAAL: 150**